

SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 1.1: Introduction to General Science — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 1.1: Introduction to General Science
Driving Question	DRIVING QUESTION: How does a scientist know something is true when they cannot see it directly — and how is this power of General Science shaping life, careers, and the environment in Kenya?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 How Did She Know? — Anchoring the Phenomenon & Opening the Driving Question Board	A photograph of a wilting maize field near Kisumu: the border plants are yellow and drooping while the interior plants are green and healthy, despite identical watering and fertilisation across the entire shamba.	Scientists — and observant Kenyan farmers — can draw reliable conclusions about things they cannot see directly by carefully reading patterns of visible evidence. This practice is called inference, and it is one of the most powerful tools in General Science. It is used daily by Kenyan professionals: soil scientists at KALRO, lab technologists at Kenyatta National Hospital, and environmental scientists monitoring Lake Victoria.	The farmer inferred a hidden soil pest because the pattern of wilting (border only, not random), the elimination of other causes (equal water and fertiliser), the time of season, and her knowledge of local pests all pointed to one hidden explanation. She did not need to see the pest — the evidence spoke for it. This is the opening puzzle of our sub-strand: How does General Science allow us to know what we cannot directly observe?
Lesson 2 What is General Science? Mapping the Discipline	Learners researched the meaning of General Science using digital devices and/or print media, and presented at least three ways General Science connects to community life in Kenya (food and agriculture, health, environment, technology).	General Science is not merely a collection of facts — it is a systematic way of asking questions and finding answers using evidence. The Kisumu maize farmer, who inferred a hidden soil pest from the pattern of wilting without seeing the pest directly, is practising General Science.	Learners co-constructed a class definition of General Science, annotated their opening predictions to show conceptual change, posted new questions on the Driving Question Board, and built Version 1 of a cumulative mind-map model that will grow across all eight lessons in this sub-strand.
Lesson 3 Importance of General Science in Life, Environment and Technology	Learners examined six Kenya-context scenario cards showing General Science at work in human health, environmental monitoring, and technology — including KEBS aflatoxin testing in maize flour,	General Science operates across three major domains of Kenyan life — human life (nutrition, disease), environment (conservation, pollution), and technology (communications, water treatment, food	The reason General Science is important is not simply that it produces useful facts — it is that it provides a systematic, evidence-based method for knowing what is real even when direct observation is impossible. The

	NEMA air-quality sensing on Mombasa Road, KWS wildlife inference in Tsavo, KNH chest X-ray reading, Safaricom M-PESA radio-wave engineering, and Nairobi Water bacterial testing. In every case, the key action was the same as the Kisumu farmer's: inferring a hidden, invisible entity from indirect evidence.	safety). In all three domains, the defining power of science is the ability to make reliable conclusions about things that cannot be directly observed, using indirect evidence and inference. This is not exceptional — it is the everyday work of scientists, doctors, engineers, and environmental officers across Kenya.	Kisumu farmer, the KEBS food-safety technologist, the NEMA engineer, and the Safaricom radio engineer are all practising the same core scientific habit: collecting indirect evidence, recognising patterns, and inferring a hidden cause. This principle — inference from indirect evidence — is the thread that connects the meaning of General Science (Lesson 2) to its importance in life, environment and technology (Lesson 3), and will next connect to the careers built on this skill (Lesson 4).
Lesson 4 Careers in General Science: Research, Chart Making & Career-Path Justification	Learners used digital devices or print media (Career Resource Sheets featuring Kenyan science professionals at KNH, KEMRI, NEMA, KMFRI, KEPHIS and county agricultural offices) to research at least four General Science-linked careers in Kenya, recording findings in a structured four-column Career Evidence Table (Career Name / Science Sub-topic / Kenyan Institution or County / How they use inference).	Learners constructed personal career-path charts mapping biology, chemistry, and physics sub-topics to specific Kenyan careers, wrote a 3-sentence career justification with a mandatory inference-connection sentence linking their chosen career back to the Kisumu maize-farmer phenomenon, and updated the Driving Question Board with a new sub-question: 'What science skills do I need for my chosen career?'	Through the cold-call share-out (five learners, 8 seconds wait time each) and the cumulative model's new Careers Branch, learners articulated that General Science careers — from agronomist in the Rift Valley to lab technologist at KEMRI to environmental scientist at NEMA — all depend on the same core scientific practice that the Kisumu farmer used: drawing reliable conclusions from indirect evidence that cannot be directly seen.
Lesson 5 Introduction to Scientific Processes and the Principle of Inference (Explain Phase – Part 1)	In three Kenyan scenarios — the Kisumu maize farmer, the Nairobi radiologist Dr. Kamau at Kenyatta National Hospital, and Amboseli Warden Lesiamon — each professional reached a reliable conclusion about something they could NOT directly see: a soil pest destroying roots, a bacterial infection in a patient's lung, and a poacher's path through the park. Learners observed that in every case, the conclusion was reached by systematically examining available physical evidence.	Inference is a conclusion drawn from available evidence combined with prior knowledge and logical reasoning — it is NOT based on direct observation of the thing being concluded, and it is NOT a guess. Scientific inference has three components: (1) Evidence — what can be directly observed or detected; (2) Reasoning — the prior knowledge and logic connecting evidence to a conclusion; (3) Inference — the conclusion about something not directly observable. The principle of inference is the foundation of all scientific knowledge because most scientific phenomena (soil pests, diseases, subatomic particles, ancient events) cannot be directly observed.	The Kisumu farmer Akinyi knew there was a soil pest destroying her maize roots — even though she never saw the pest — because she applied the principle of inference: her evidence (wilting and yellowing confined to one edge, uniform watering and fertilising, dry season timing, no visible above-ground pest) combined with her reasoning (soil pests are seasonally active in dry periods and cause localised root damage that produces exactly this pattern of wilting) led her to a reliable inference (a soil pest is destroying the roots underground). This is not a guess — it is science.
Lesson 6 Collecting Evidence and Drawing Inferences: Mystery Bags Hands-On	Learners opened sealed mystery evidence bags containing written clue cards describing plants in Kenyan farming contexts (Kisumu, Rift Valley, Kiambu,	Learners identified and named the methods scientists use to collect evidence (observation, measurement, experimentation, record-keeping) and	Learners explained, using the sentence frame 'Based on [evidence] and [evidence], our group infers [cause] because [reasoning],' why indirect evidence —

Activity	Mombasa, Baringo). They read, sorted, and classified written evidence clues — no images, no direct plant observation — into categories (pest / disease / nutrient-environmental problem) using an Evidence Classification Table.	mapped their own group actions onto the KICD science process skills: classifying, hypothesising, identifying variables, interpreting data, and drawing conclusions. They distinguished between an observation and an inference, and recognised that contradictory evidence must be weighed — not ignored — for an inference to be reliable.	systematically collected and classified — is sufficient for a scientist to draw a reliable conclusion. They connected this to the anchoring phenomenon: the Kisumu maize farmer used the same evidence-collection and inference process they practised today, demonstrating that General Science process skills are tools used in real Kenyan agricultural and professional contexts every day.
Lesson 7 DQB Curation and Cumulative Model Building	Students observed the full Driving Question Board from Lessons 1–6 and sorted all question cards into three columns — 'We Can Now Answer This,' 'We Partly Understand This,' and 'We Still Don't Know This' — justifying each placement with specific evidence from previous lessons.	Students learned that scientific understanding is cumulative and incomplete — some questions from their DQB can now be answered with evidence, others require more investigation in future sub-strands (The Cell, Nutrition, Respiration, etc.), and the principle of inference connects General Science to careers across Kenya, from doctors at Kenyatta National Hospital to fisheries officers at Lake Victoria to agronomists in Kericho.	Students explained, through their Cumulative Science Models and peer-reviewed mind-maps, how the Kisumu maize farmer's act of inferring a hidden soil pest from indirect evidence is the same cognitive move used by scientists and professionals across Kenya — and how the entirety of General Science is built on this principle of making reliable conclusions from evidence that cannot always be directly observed.
Lesson 8 Synthesis, Presentations and Unit Assessment — Final Explanation & Model Consolidation	Learners observed all peer groups present their career-inference charts, recording in a structured listening frame how each career — from KALRO agronomists to KNH lab technologists to Kericho Tea Research Institute scientists — uses indirect evidence and inference as a core professional practice. They observed the class Driving Question Board being colour-coded for the first time: green ticks for questions fully answered, yellow arrows for partial answers, and red question marks for questions (especially about the cell and root destruction) that remain open and will anchor Sub-Strand 1.2.	Learners learned that the principle of inference — drawing a reliable conclusion about something not directly observable, using evidence and prior knowledge — is not a single lesson topic but the unifying thread of the entire sub-strand and of General Science as a discipline. They learned that a new but structurally identical Kenyan scenario (the Kericho tea farmer's curling leaves) can be analysed using the same three-part framework: identify evidence, state an inference about a hidden cause, and name a career that would investigate it professionally. They also learned that their understanding has grown measurably since Lesson 1 — visible in the annotated comparison of their initial and final models.	The final anchor-chart answer co-constructed by the class directly explains the anchoring phenomenon: the Kisumu maize farmer knew a soil pest was destroying her roots without seeing it because she applied the principle of inference — she identified a pattern of indirect evidence (localised wilting, yellowing, absence of other causes, season of year), used her prior knowledge of maize pests, and drew a reliable conclusion about a hidden cause. This is not unique to farmers — it is the daily practice of every General Science professional in Kenya, and it is the foundation of everything the class will study in Sub-Strands 1.2 through 3.4.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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